Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/09/2007
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Project Title:	Effect Of Light On Plant Gravitropism: Functional Genomic Analysis		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Lomax, Terri Ph.D. (Oregon State University) Fowler, John (Oregon State University)		
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Task Description:	The response of higher plants to gravity is controlled by integrated interactions among environmental and endogenous signals. Interactions between light and the gravitropic response have long been recognized, but the underlying mechanism is poorly understood. Fundamental knowledge in this field will allow effective application of light to improve plant growth in space, which is essential biological knowledge to support NASA's goals for long-duration human presence and exploration in space. Tomato plants carrying a mutation in the Lazy-2 gene exhibit a reversed gravitropic response under red light and grow downward instead of upward. Further study of the lazy-2 tomato mutant provides a unique opportunity to understand the mechanism that permits light to modulate the gravitational set-point angle of shoots. To our knowledge, the lazy-2 lesion is the only mutation that reverses the directionality of the gravitropic response rather than reducing or eliminating it. Our goal is to utilize the lazy-2 lesion to elucidate the molecular mechanisms underlying light regulation of gravicurvature. To this end, we plan to 1) use auxin-responsive gene constructs as reporters to study changes in auxin transport and sensitivity, 2) identify molecular components from the Lazy-2 signal transduction pathway via global expression profiling, 3) clone the Lazy-2 gene to identify its role in light regulation of the gravitropic response, and 4) perform comparative studies of the phylogeny of the Lazy-2 gene and other genes involved in light/gravity interactions. The proposed research supports the goals of NASA's Fundamental Space Biology Program in determining the role of gravity on normal plant development; developing an understanding at the genetic, molecular, and cellular levels of how the expression of genes is controlled by the force of gravity and light; and establishing how the mechanisms by which plants perceive, transduce, integrate, and respond to gravity and red light compare among plant species. Tomato plants carrying a mutation in the Lazy-2 gene display a striking phenotype. Under red light, lazy-2 mutants exhibit a reversed gravitropic response and grow downward instead of upward. In darkness or blue light, the mutants assume the upright growth habit characteristic of wild-type plants, indicating that the Lazy-2 gene product acts in a branch of the shoot gravitropic pathway that is specifically regulated by red light. Our goal is to utilize the lazy-2 lesion to elucidate the molecular mechanisms underlying light regulation of gravicurvature. The aim of this project is to enhance our understanding of how the combined forces of gravity and light interact to regulate plant development, which will lead to the discovery of mechanisms that insure or refine plant production in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Under the current support, the following received training: Postdoctoral Fellows: Kwangchul Oh, Warren Coffeen Graduate Students: Kirk Findlay Ph.D Botany and Plant Pathology student at OSU (June 2003-December 2004, interrupted). The project involved molecular and cellular analysis of the mechanisms underlying the reversed gravitropic response in the tomato mutant lazy-2. Udergraduate Students: Katherine VanWormer, College of Honor Botany and Plant Pathology & Chemistry student at OSU (June 2004-May 2005). Mis VanWormer defended a thesis on her project. The project involved experimental manipulations directed to phenocopy, or partially restore, the light-mediated reversed gravitropic response in the tomato mutant lazy-2. Will Shappy, undergraduate researcher (September 2002-October 2003). The project involved screening of populations of recombinant plants to precisely map the Lazy-2 gene in tomato. Anthony Mengucci, undergraduate researcher (January 2001-Desember 2003). The project involved genotyping of populations of recombinant plants to precisely map the Lazy-2 gene in tomato. Jay Well, College of Honor Botany and Plant Pathology student at OSU (2001-2003). Mr. Well defended a thesis on his project). The project involved a genetic screen as a part of the map-based cloning of the lazy-2 gene in tomato).
	During the granting period, we have addressed two specific objectives within our research strategy: (1) high-resolution mapping and molecular cloning of the Lazy-2 gene, and (2) uncovering the possible role of auxin in the reversed gravitropism of lazy-2 under red light. Using a mapping population of plants generated from the cross between L. esculentum and L. pimpinellifolium, we have made considerable progress toward the fine mapping the Lazy-2 gene on chromosome. During the granting period, we screened over 6,000 recombinants from the above genetic cross, and identified over 40 plants with recombinations between molecular markers TG504 and CT201, close to Lazy-2 on chromosome 5. Polymorphisms were identified between L. esculentum and L. pimpenelifolium sequences, allowing for their mapping within this region. These analyses allowed us to place eight markers (TM5, TG619, T0998, T1174, TG344, TG363, TG448, TG606) within the genetic interval defined by Lazy-2, between markers TG504 and CT201 on chromosome 5. Eight of the identified recombinants provided information for further separation of this region into five zones. In this mapping, the Lazy-2 gene was localized 0.04 cM from marker TG344 and 0.07 cM from marker T1174. With the resolution of the Lazy-2 gene into a small genetic region, we attempted localization of the gene on a contiguous architecture of overlapping sequences (contig). We screened a tomato L. cheesemanii BAC genomic DNA library (LA483, courtesy of Dr. J. Giovannoni, Cornell University) that represents approximately 15 haploid genome equivalents arrayed on nylon membranes, providing a greater than 99% probability of representing any specific tomato sequence. Using marker TG344 as a labeled probe, we identified two BAC clones that hybridized to this sequence. The ends of the BAC clone inserts were sequenced from the adjacent SP6 and T7 primer sites located on the cloning vector. Specific products were generated via PCR for each BAC end sequence and used as labeled probes against digested BAC DNAs in order to orient the clones with respect to each other, to eliminate false-positives, and to identify the two outmost ends of the resulting contig. Using the same approach in three more cycles of contig assembly has extended the contig to at least 200-300 kb length.Polymorphisms between L. esculentum and L. pimpenellifolium were identified for four of the BAC-end sequences from the contig, allowing conversion of these sequences into four new molecular markers, 164C5Sp6, 184M24T7, 213B16T7, 121L18T7. All four new markers generated from the contig co-segregated with marker TG344, and not with the lazy-2 phenotype, demonstrating that the contig had not reached the location of the Lazy-2 gene yet. In a complementary approach to identify new molecular markers, we performed searches for conserved synteny (co-linearity) of genes from the Lazy-2 region on chromosome 5 in tomato, and the chromosomes of the reference plant Arabidopsis where the entire genome sequence is available. Conserved synteny allows to perform educated predictions of gene linearity in one organism based the gene linearity in another organism. Such strategies allowed us to map the

Task Progress:	Diageotropica locus of tomato into a chromosome region of only 30 kb on chromosome 1 of tomato (Oh et al., 2002). We performed homology searches of Arabidopsis databases with tomato sequences of molecular markers that we have identified as being closely linked to Lazy-2 (Fig.1), as well as tomato RFLP and cosmid markers mapping close to this region (in the Solanaceae Genomics Network, http://www.sgn.cornell.edu). Tomato sequences were blasted (tblastx) against genomic and EST-containing Arabidopsis databases (http://www.arabidopsis.org). This analysis revealed synteny between the Lazy-2 region and an area on Arabidopsis chromosome 1: This area contained sequences homologous to seven tomato markers spanning the entire Lazy-2 regionregion (from marker TM5 to marker TG448). Representative genes from the Arabidopsis BAC clones can be used to identify orthologous tomato genes (at http://www.sgn.cornell.edu), that can be tested for proximity to Lazy-2. Based on our analysis, Arabidopsis genes between Atg67730.1 and Atg68610.1 are most likely to provide informative markers within the lazy-2 region. We have also performed analysis aimed to uncover the nature of the protein encoded by Lazy-2 gene. The plant hormone auxin plays a major role in the transduction of gravitropic signals from cell to cell, as well as in the control of differential growth that follows gravistimulation. To explore possible abnormalities in auxin signal transduction during the gravicurvature of the lazy-2 mutant, we cloned the promoters of the auxin-response tomato genes, LeIAA3 and LeIAA11, in front of the reporter GUS gene. The resulting constructs, together with a highly active synthetic Arabidopsis auxin response promoter, DR5 (courtesy of G. Haggen, Dept. Biochemistry, U. Missouri), were stably transformed in tomato via Agrobacterium mediated transformation. We found that expression of LeIAA3 and 11 was too low to allow for detection upon gravistimulation. However, expression of the DR5 reporter permitted detection upon gravistimulation. We observed that expression of DR5:GUS was reversed in lazy-2 upon gravistimulation under red light, so that activity of the GUS protein was observed on the upper side of the hypocotyl. This was in contrast to the wild-type plants from the same population, which expressed the reporter on the lower gravistimulated side, as expected. This observation supports that in gravistimulated lazy-2 plants the lateral gradient of auxin is reversed, leading to auxin accumulation on the upper instead on the lower side of the gravistimulated hypocotyls. Using the same population of transgenic plants, we also compared the intensity of GUS activity between wild type and lazy-2 plants. This comparison revealed that the level of DR5:GUS expression was generally reduced in lazy-2 plants in comparison to wild-type plants from the same population. Thus, the lazy-2 mutant, in addition to exhibiting a reversed gradient of auxin upon gravistimulation, also exhibits deficiency in either the level of auxin, or the level of auxin response upon gravistimulation. In addition, we performed treatments of tomato seedlings with chemicals known to affect auxin signaling and gravitropism in wild-type plants. We observed that the auxin-transport inhibitor naphthalene-phtalamic acid, NPA, affected the downward curvature of lazy-2 plants in a manner similar to its effect on the upward curvature of wild-type plants: In both cases, an inhibition of the curvature was observed. This supports that auxin transport is involved in generation of the reversed gravicurvature of the mutant. The phase of the gravitropic response upstream of auxin transport is likely to involve processes within the actin cytoskeleton. Latrunculin B, a potent actin cytoskeleton-disrupting compound, caused an enhancement of gravicurvature of wild type tomato seedlings, while shoot elongation was reduced. Contrary to wild type, lazy-2 seedlings gravistimulated under red light in the presence of Latrunculin B appeared to curve erratically, as if they had lost the ability to either perceive or adjust to gravity. This suggests that alterations in the ability of the actin cytoskeleton to perceive gravity might also account for the generation of the lazy-2 mutant phenotype. Thus, our analysis suggests that the Lazy-2 gene product acts early in the gravitropic response, likely during the poorly understood phase of the pathway between perception of gravity and lateral redistribution of auxin, which involves processes within the actin cytosceleton.

Bibliography Type:	Description: (Last Updated: 04/10/2007)
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